Hardware Design Reference

Integrated Automated Feeding & Watering System with Centralized Data Monitoring for Commercial Piggeries

Circuit, wiring & connection diagrams derived from the thesis manuscript and the hardware connections & wiring guide.

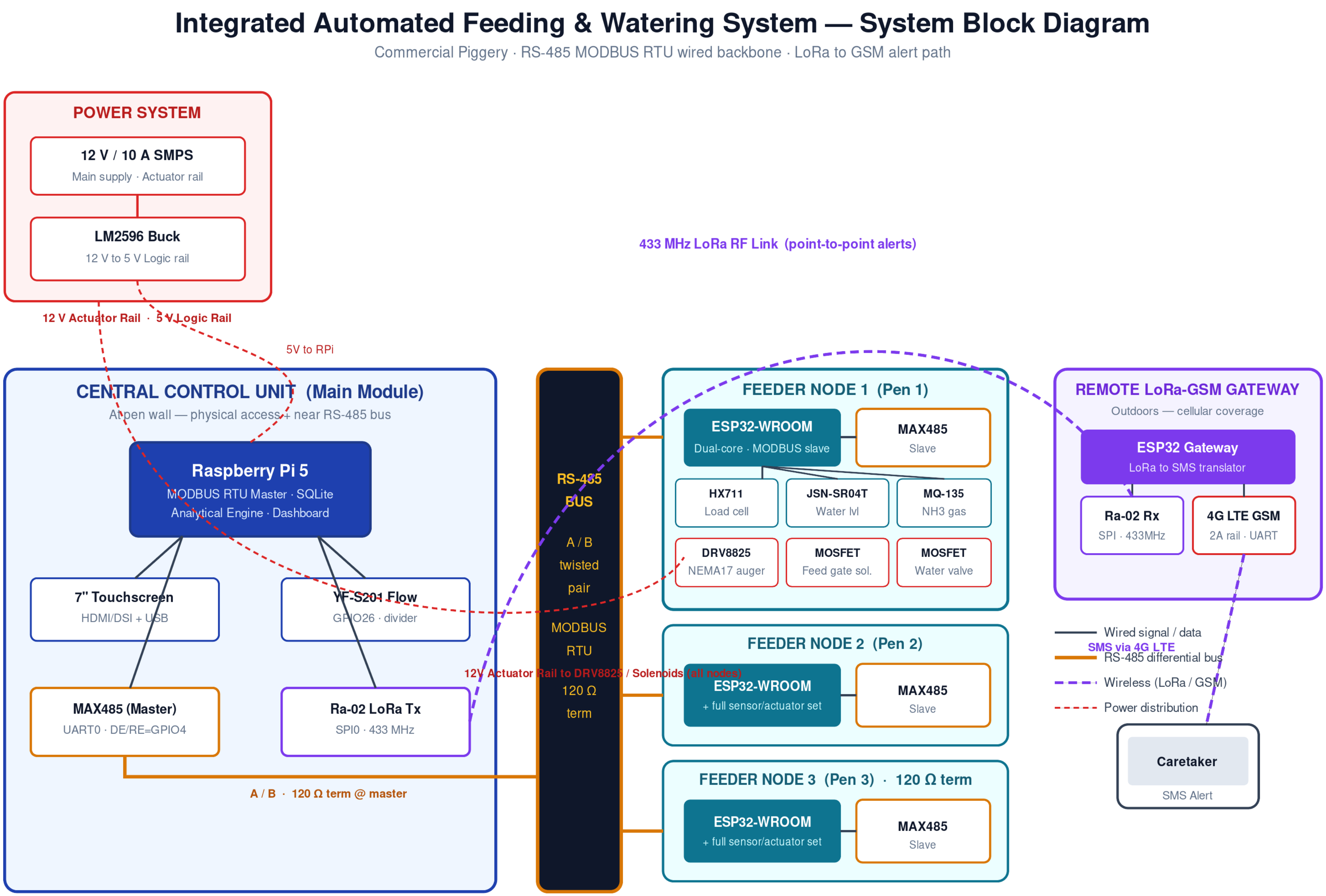
Raspberry Pi 5 MODBUS RTU master · 3 x ESP32-WROOM-32 feeder nodes
RS-485 wired backbone · LoRa 433 MHz · 4G LTE-GSM SMS gateway

Contents:

Followed by pin-mapping reference tables.

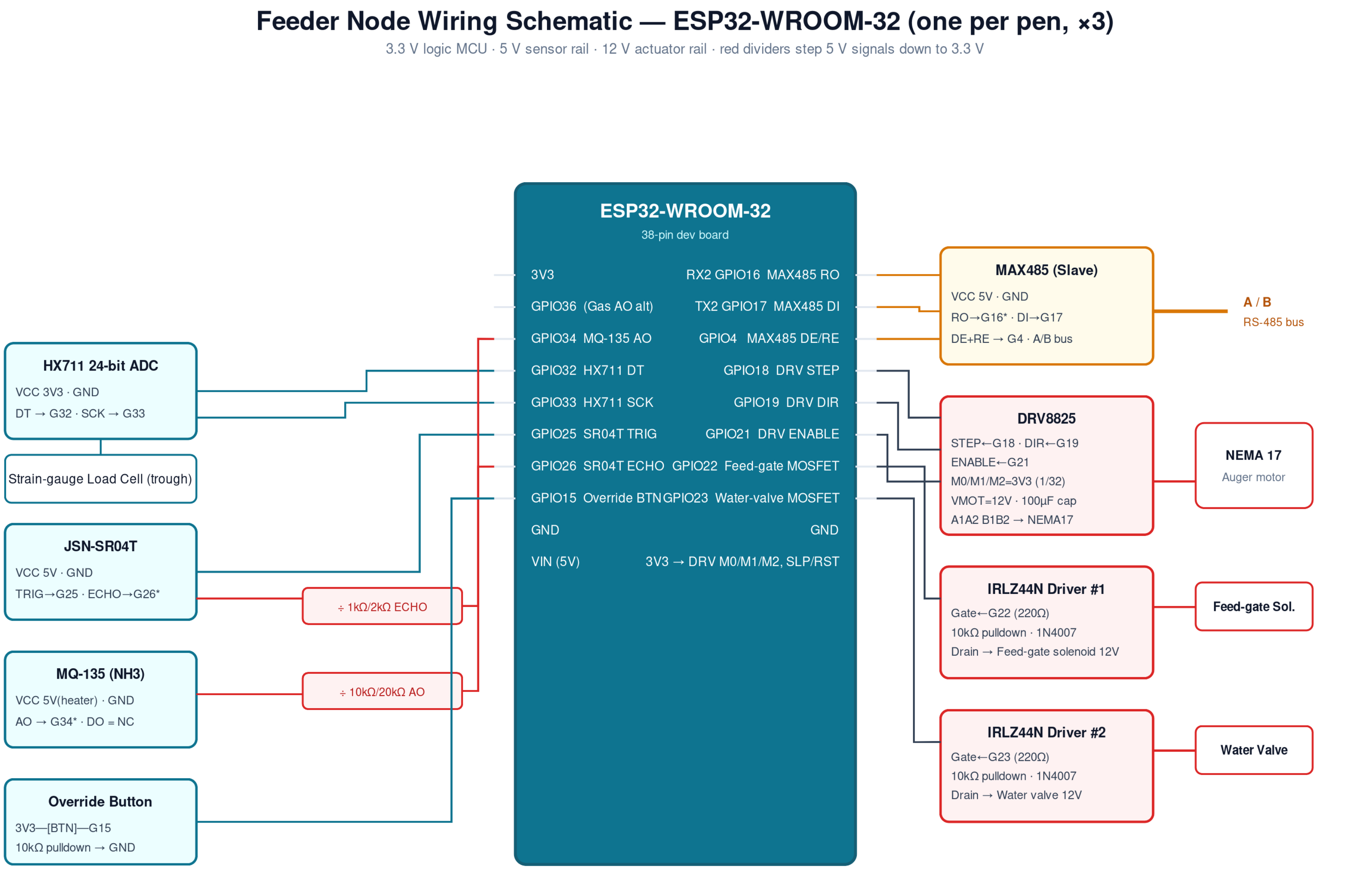
1 · System Block Diagram

Overall topology, power rails, RS-485 bus, feeder nodes, LoRa/GSM alert path



2 · Feeder Node Wiring Schematic (ESP32-WROOM-32, x3)

Per-pen node: sensors, DRV8825 stepper, MOSFET solenoid drivers, MAX485 slave



* Red blocks = resistor voltage dividers protecting 3.3 V ESP32 GPIO from 5 V sensor outputs (ECHO, MQ-135 AO, MAX485 RO)

Node power & ground

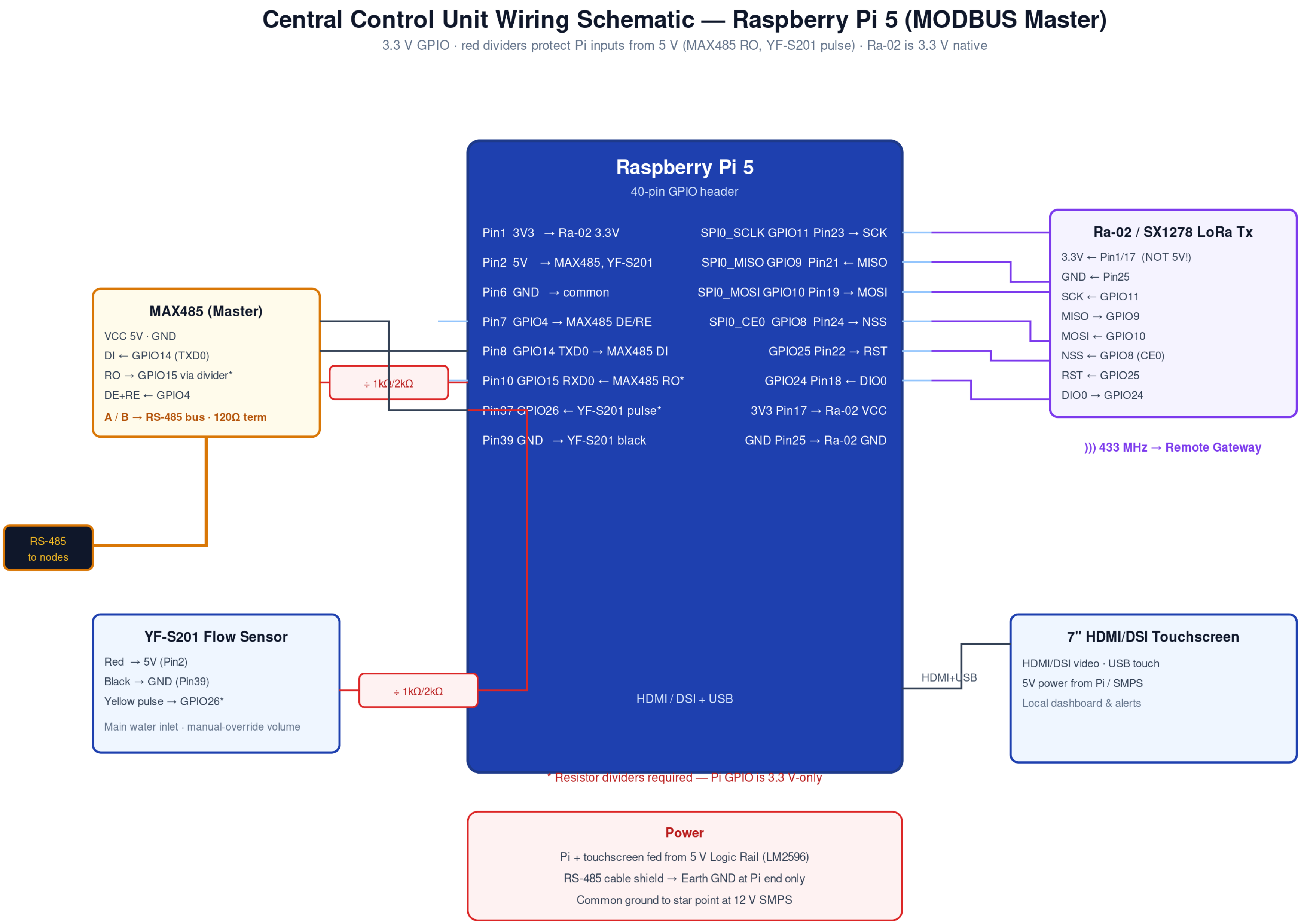
12 V Actuator Rail → DRV8825 VMOT, solenoid coils

5 V Logic Rail (LM2596) → ESP32 VIN, MAX485, JSN-SR04T, MQ-135 heater

Common GND (star point at SMPS) · HX711 & ESP32 3V3 from onboard regulator

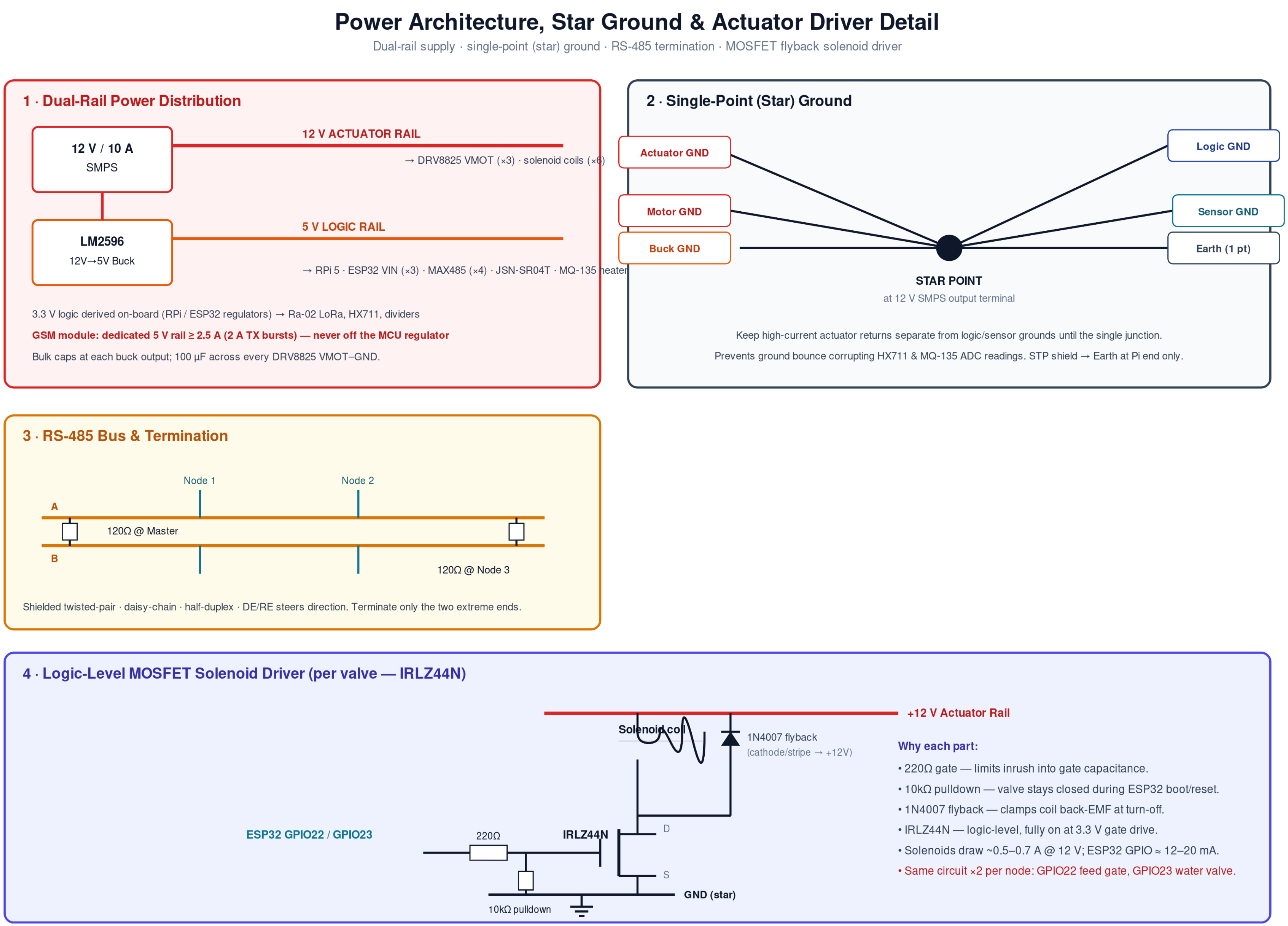
3 · Central Control Unit Wiring Schematic (Raspberry Pi 5)

MODBUS master, Ra-02 LoRa (SPI0), YF-S201 flow, touchscreen, level shifters



4 · Power Architecture, Star Ground & Driver Detail

Dual-rail supply, single-point ground, RS-485 termination, MOSFET flyback driver



Reference A · Central Unit (Raspberry Pi 5) Pin Map

Master controller — 3.3 V GPIO, dividers required on 5 V inputs

Peripheral / Pin	RPi 5 pin	GPIO	Notes
MAX485 VCC	Pin2 / 4	5V	Transceiver power
MAX485 GND	Pin6	GND	Common ground
MAX485 RO -> Pi RX	Pin10	GPIO15 (RXD0)	1k/2k divider (5V->3.3V)
MAX485 DI <- Pi TX	Pin8	GPIO14 (TXD0)	Direct 3.3V->5V OK
MAX485 DE+RE	Pin7	GPIO4	Tied together; High=TX
Ra-02 VCC	Pin1 / 17	3.3V	Do NOT use 5V
Ra-02 SCK	Pin23	GPIO11 (SCLK)	HW SPI0
Ra-02 MISO	Pin21	GPIO9	HW SPI0
Ra-02 MOSI	Pin19	GPIO10	HW SPI0
Ra-02 NSS	Pin24	GPIO8 (CE0)	Chip select
Ra-02 RST	Pin22	GPIO25	GPIO reset
Ra-02 DIO0	Pin18	GPIO24	TX-done / RX IRQ
YF-S201 Red	Pin2	5V	Sensor power
YF-S201 Black	Pin39	GND	Ground
YF-S201 Yellow	Pin37	GPIO26	1k/2k divider on pulse
Touchscreen	HDMI/DSI+USB	-	5V powered

Reference B · Feeder Node (ESP32-WROOM-32) Pin Map

One identical node per pen (x3) — MODBUS RTU slave

Peripheral / Signal	ESP32 pin	GPIO	Notes
MAX485 RO	RX2	GPIO16	1k/2k divider (5V->3.3V)
MAX485 DI	TX2	GPIO17	Direct
MAX485 DE+RE	G4	GPIO4	Direction select
HX711 DT	G32	GPIO32	Load-cell data (in)
HX711 SCK	G33	GPIO33	Load-cell clock (out); VCC=3.3V
JSN-SR04T TRIG	G25	GPIO25	Trigger pulse; VCC=5V
JSN-SR04T ECHO	G26	GPIO26	1k/2k divider on echo
MQ-135 AO	G34	GPIO34	10k/20k divider; VCC=5V heater
Override button	G15	GPIO15	3.3V + 10k pulldown, active-high
DRV8825 STEP	G18	GPIO18	Step pulse
DRV8825 DIR	G19	GPIO19	Direction
DRV8825 ENABLE	G21	GPIO21	Low=enabled
DRV8825 M0/M1/M2	3.3V	-	1/32 microstep; RST+SLP->3.3V
DRV8825 VMOT	12V rail	-	100uF cap VMOT-GND; A1A2B1B2->NEMA17
Feed-gate MOSFET	G22	GPIO22	IRLZ44N; 220R gate, 10k pd, 1N4007
Water-valve MOSFET	G23	GPIO23	IRLZ44N; 220R gate, 10k pd, 1N4007
ESP32 power	VIN	-	5V logic rail (LM2596)

Reference C · Remote LoRa-GSM Gateway (ESP32) Pin Map

Outdoor unit — receives LoRa, sends SMS via 4G LTE

Peripheral / Signal	ESP32 pin	GPIO	Notes
Ra-02 SCK	G18	GPIO18 (VSPI)	HW SPI
Ra-02 MISO	G19	GPIO19	HW SPI
Ra-02 MOSI	G23	GPIO23	HW SPI
Ra-02 NSS	G5	GPIO5	Chip select
Ra-02 RST	G14	GPIO14	Reset
Ra-02 DIO0	G2	GPIO2	RX packet IRQ
GSM TXD -> ESP RX	RX2	GPIO16	Serial from module
GSM RXD <- ESP TX	TX2	GPIO17	Serial to module
GSM PWRKEY	G12	GPIO12	Pulse low 1.5 s to boot
GSM VCC	5V rail >=2.5A	-	2A TX bursts; dedicated rail

Reference D · Power & Noise-Isolation Rules

Assembly rules for a hostile (high-EMI) piggery environment

1. Dual-rail power: 12 V/10 A SMPS actuator rail feeds DRV8825 VMOT and 12 V solenoid coils; LM2596 buck derives the 5 V logic rail for MCUs, sensors and transceivers.
2. Single-point (star) ground: keep actuator returns separate from logic/sensor grounds until a single junction at the 12 V SMPS output terminal. Prevents ground bounce corrupting HX711 & MQ-135 ADCs.
3. RS-485 bus: shielded twisted pair, daisy-chain, half-duplex. Shield to Earth at the Pi end ONLY. 120 ohm termination across A/B at the two extreme ends (master MAX485 and Node-3 MAX485).
4. Level shifting: all 5 V signals into 3.3 V MCU inputs use resistor dividers (MAX485 RO, JSN echo, MQ-135 AO, YF-S201 pulse).
5. Transient protection: 100 uF across every DRV8825 VMOT-GND; 1N4007 flyback across every solenoid; 220 ohm gate resistor + 10k gate pulldown on every MOSFET.
6. GSM power: dedicated 5 V rail capable of >=2.5 A (2 A TX bursts) — never off the MCU regulator.
7. Cable routing: run sensor cables (HX711, JSN-SR04T) in separate conduits away from 12 V actuator wiring.